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Substantiation of the Geometric and Operating Parameters of Shearer Auger Drums Ensuring Efficient Loading of Coal onto the Face Conveyor

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Abstract

The processes of unloading coal from the cutting (breakage) zone and loading it onto the face conveyor by the auger drums of shearers in fully mechanized longwall faces of coal mines are analysed. Possibilities for increasing the efficiency of coal loading onto the face conveyor are considered with respect to the criteria of reducing additional comminution of coal and specific energy consumption, which is achieved by increasing the cross-sectional area of the loading window and the volume of the loading zone on the conveyor.

Keywords: *underground coal mining, fully mechanized longwall face, shearer, auger drum, face conveyor, coal, loading, comminution, efficiency.*

Introduction

In mechanized longwall systems, the auger drum of a shearer performs the main function: separating coal from the solid seam and loading it onto the face conveyor. Auger drums are distinguished by their technological effectiveness, compactness, simplicity of design, reliability and durability [1, 2, 3]. However, auger drums trailing in the direction of shearer travel are characterized by unsatisfactory floor clean-up, which leads to a reduction in shearer productivity, an increase in specific energy consumption and additional comminution of coal as it is loaded onto the conveyor. The main reasons for this situation are the limited size of the loading window and the large distance between the drum and the conveyor [1, 4, 5]. The problem is that there are no technical solutions or proven methods for determining the structure and parameters of shearer auger drums and their fit, together with the conveyor and loading shields, within the face space while achieving the required values of the efficiency indicators of the process of loading broken coal onto the face conveyor.

Substantiation of the parameters of shearer auger drums

Investigating the loading process and substantiating the parameters of shearer auger drums that ensure an increase in the efficiency of the loading and conveying process is a topical scientific problem.

The movement of bulk material in horizontal and slightly inclined screw loading and conveying mechanisms with a relatively low fill factor of the working space (up to 0.3–0.4) is accompanied by the formation of a drag body in front of each flight. Inside the drag body, circulation processes take place, caused by the differently directed forces of gravity, inertia and friction [6, 7]. This causes the main drawback of screw conveyors: increased energy consumption for moving the material and its

additional, secondary (often undesirable) comminution. The intensity of circulation of the conveyed material within the volume of the drag body increases sharply when a certain critical deflection angle $\theta_к$ of the drag body (Fig. 1) in the direction of drum rotation is reached, at which conditions are created for the material to slide down along the line of least resistance on its free surface.

[Figure 1 — see original PDF, page 2] Fig. 1. Mass movement in an auger

The deflection angle of the material can be determined by the expression [6]:

$$\theta_к = arcsin\ (\ (tg\ \beta + f_2 R\omega^2 / (g\ cos\ \beta)) / \sqrt{[\ (2\pi R / S) \cdot (1 - f_1\sqrt{(ctg^2\gamma + (S / 2\pi R)^2 + 1) + f_2\ ctg\ \gamma) + f_2^2\ ]}\ ) + arctg\ (\ f_2 / [\ (2\pi R / S) \cdot (1 - f_1\sqrt{(ctg^2\gamma + (S / 2\pi R)^2 + 1) + f_2\ ctg\ \gamma)\ ]\ ),\ (1)$$

where: γ is the angle of inclination of the drum axis to the horizontal plane; β is the angle of inclination of the drum axis to the horizontal plane; R is the radius of the guiding surface, m; S is the pitch of the drum helix, m; f_1 is the coefficient of friction of the material against the surface bounding the flow (the face surface, retaining shield, guide tube, borehole); f_2 is the coefficient of friction of the moving material against the flight; ω is the rotational speed of the drum, 1/s; g is the acceleration due to gravity, m/s^2.

[Figure 2 — see original PDF, page 2] Fig. 2. Dependence of the output of the auger drum in terms of broken coal $Q_{тр}$ and circulating coal $Q_ц$ as a function of the fill factor of the drum cross-section [1]

The degree and nature of the influence of the geometric and operating parameters of the drum and of the properties of the conveyed material on the value of θ_K are presented in the studies of M.S. Safokhin, I.M. Pitbaum, V.N. Verner and others [5, 6, 8]. Taking into account the dependences given, one can judge the equivalence of the influence of various factors on the deflection angle of the material movement in the drum along lines of identical regimes. The greater the deflection angle of the material θ_K, the higher the circulation intensity $k_ц$.

The filling of the drum cross-section depends on the difference between the diameters of the drum and its hub. As indicated in [1, 6], increasing the filling of the drum leads to an increase in the circulation coefficient. The values of the fill factor $\psi_н$ and the circulation intensity $k_ц$ depend substantially on the haulage speed of the shearer and the drum diameter. In addition, the circulation intensity depends on the throwing capacity of the drum, which increases with increasing helix angle, since the force components p_r and p_θ (see Fig. 1) that throw the material increase with increasing helix angle α_i. The curve has a minimum at values α_i = 18–23° [1].

The intensity of material circulation during loading can be estimated by the circulation coefficient $k_ц$ for any value of the fill factor $\psi_н$:

$$k_ц = Q_ц / (Q_{mp} - Q_ц),\ (2)$$

where: $Q_{тр}$ is the output of the auger drum in terms of cut coal; $Q_ц$ is the output of the drum in terms of circulating material (Fig. 2).

Line 1 (see Fig. 2) characterizes the dependence of the drum output in terms of broken coal, and line 2 the volume of coal in terms of circulating material. The value $\psi_{н\ max}$ determines the value of the fill factor above which exceeding the critical value leads to an increase in the volume of circulating material.

Depending on the direction of rotation of the trailing auger drum of the shearer, the broken material is arranged inside the drum space in a certain way (Fig. 3).

With co-directional rotation of the drum (from floor to roof) relative to the shearer haulage speed (see Fig. 3, a), the bulk of the broken coal falls down and is fed by the drum onto the face conveyor, remaining in the face-side working space. In this case, coal comminution is reduced. The angle φ (see Fig. 3) depends on the coefficient of friction of coal against the surface of the drum flights and hub and is proportional to the deflection angle of the material position θ_K. As the drum output increases, its fill level at the outlet increases (see Fig. 3, positions I, II).

With counter-rotation of the drum (from roof to floor) relative to the haulage speed (see Fig. 2, b), the bulk of the coal being loaded is located on the side of the loading shield, forming an angle φ with the seam floor. As the shearer haulage speed increases, the coal flow in the drum increases. The flow velocity at constant values of rotational speed, number of starts and lead of the drum will remain unchanged. A moment is possible when the material will roll onto the drum barrel and be thrown over into the zone of the clean-up plate. Here it either remains on the seam floor in the absence of a clean-up plate (Fig. 3, c), increasing the height of the unloaded layer H_0 [1], or is gathered by the moving plate and fed by the drum onto the conveyor, being partially carried into the zone of the drum working space and forming a flow of circulating material.

[Figure 3 — see original PDF, page 3] Fig. 3. Diagrams of the arrangement of broken coal in the working space of the trailing drum

Fig. 3 shows that the coal flow is not unloaded over the entire cross-sectional area of the drum. The zone of its unloading depends on the geometric and operating parameters of the drum. In actual designs, the area of the loading window in the unloading zone is limited by the diameter of the drum hub, the height of the face conveyor side and the dimensions of the drum gearbox housing. The limited size of the window increases the resistance to the movement of the coal flow, which leads to an increase in the fill factor of the drum.

With counter-rotation (see Fig. 3, a, c) and increasing haulage speed, the area of the coal flow located in the working space of the drum increases [10, 11, 12]. An increase in the flow height leads to increased circulation of coal in the working space of the auger drum. On closer examination, the loading process as a whole, in its physical essence, can be represented by three different processes taking place in adjacent zones I, II, III (Fig. 4).

[Figure 4 — see original PDF, page 3] Fig. 4. Fill factor of the drum cross-section along the web width B_3

The values of the fill factor and the circulation intensity of the material in the drum vary along the web width B_3 (see Fig. 4). In the corner zone I, the flow of coal separated from the solid seam is small, no drag body is formed, and throwing, circulation and comminution of coal are increased. In zone II (the zone where the solid seam is broken by the face picks), the fill factor of the drum increases and a drag body forms. In zone III (unloading of coal onto the conveyor), the fill factor of the drum increases owing to back-pressure of the flow caused by the limited throughput capacity of the loading window.

With co-directional rotation of the drum, the flow is located between the floor and the shield (see Fig. 3, b). For equal drum outputs in terms of cut coal, the flow height in this case is lower than with counter-movement, and the circulation intensity decreases. At the same time, an increase in the flow area reduces its velocity.

Conclusion

Based on the above, the following conclusions can be drawn:

- the process of loading coal onto the conveyor has not been sufficiently studied; the substantial energy costs, the incidental comminution of coal and dust generation have a negative effect on the efficiency of the mining process;
- the investigated process of moving coal onto the conveyor by the auger drum should be regarded as consisting of three interrelated processes: unloading of coal from the corner zone; transport of coal by the flights along the drum across the web width; and pushing of the bulk load through the loading window onto the conveyor;
- research is needed on the influence of the drum fill factor and the size of the loading window area on the intensity of coal circulation and the efficiency of the loading process;
- it is necessary to search for and develop technical solutions that reduce the intensity of circulation and comminution of coal and reduce the resistance to its movement by enlarging the loading window.

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